Mohamed khalil **KHLIFI**

Mechatronics Engineer | AI & Computer Vision

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SUMMARY

Mechatronics Engineer Mechatronics Engineer with hands-on experience developing AI-powered computer vision applications for object detection, tracking, recognition, and image/video analysis. Experienced in integrating deep learning models, image processing, and embedded hardware into intelligent systems.

PROFESSIONAL EXPERIENCE

Final Year Engineering Intern — Neuralbey — Monastir, Tunisia | March 2025 – September 2025 | View Source Code

- Architected an IoT-based smart pet feeder featuring automated dispensing, remote monitoring and local user interface and safety features.
- Programmed the ESP32 firmware to integrate four sensors, an RTC module, stepper motor control and Blynk-based mobile communication.
- Incorporated Wi-Fi connectivity through the ESP32 to enable remote monitoring and scheduled feeding via the Blynk IoT platform.
- Laid out a custom PCB with power management circuitry and hardware interfaces for reliable system integration.

Key Technologies: ESP32 – Embedded C – IoT – Arduino IDE – PCB Design – Altium Designer – Wokwi – SolidWorks – Blynk – Sensors

Summer Engineering Intern — TAC–TIC — Ariana, Tunisia | July 2024 – September 2024 | View Source Code

- Created an AI-powered object-counting application capable of processing images, videos, and live webcam streams using YOLOv8.
- Designed a graphical user interface using Tkinter for mode selection and input-source management.

Key Technologies: Python – PyTorch – OpenCV – YOLOv8 – Numpy – PyCharm – Ultralytics – Tkinter – Computer Vision

Summer Engineering Intern — Relays and Industrial Automation (RAI) — Ariana, Tunisia | July 2023 – August 2023

- Analyzed industrial control schematics and supported component selection, electrical assembly, and PCB integration according to system requirements.
- Performed functional testing and troubleshooting to validate hardware operation and ensure system reliability.

Key Technologies: Industrial Automation – Electrical Schematics – Assembly – Functional Testing – PCB Assembly – Troubleshooting

PROJECTS

Automatic Number Plate Recognition (ANPR) | December 2024 | View Source Code

- Designed an AI-based ANPR for vehicle plate detection and recognition, supporting real-time webcam and static image processing.
- Combined YOLOv3 for license plate detection, image segmentation techniques, and a CNN-based Optical Character Recognition (OCR) model for character recognition into a unified recognition pipeline from images and live video streams.

Key Technologies: Python – YOLOv3 – OpenCV – TensorFlow/Keras – CNN – OCR – Computer Vision – Image Processing – Jupyter Notebook

YOLOv8 Multi-Object Tracking | August 2024 | View Source Code

- Developed a deep learning-based object detection and tracking system using YOLOv8 for identifying and following multiple objects in video streams.
- Built object tracking using the ByteTrack algorithm with unique object IDs for continuous monitoring across video frames.

Key Technologies: Python – YOLOv8 – OpenCV – Ultralytics – ByteTrack – Computer Vision

End of Year Project: Surveillance Robot | February 2024 – May 2024 | View Source Code

- Developed an ESP32-CAM-based system for wireless video streaming to a host computer running YOLOv3-based object detection and tracking.
- Extended remote monitoring to an existing mobile streaming application for off-site use.

Key Technologies: ESP32-CAM – Arduino IDE – SolidWorks – Fritzing – YOLOv3 – Robotics

EDUCATION

National Engineering School of Carthage (ENICarthage) | September 2021 – October 2025

- National Engineering Degree in Mechatronics Engineering (Master's equivalent)

Preparatory Institute for Engineering Studies of Sfax (IPEIS) | September 2019 – June 2021

- Engineering Preparatory Cycle – Physics and Technology

TECHNICAL SKILLS

Programming Languages: Python – Embedded C
AI & Computer Vision: Deep Learning – Object Detection – Object Tracking – Image Processing – Image Segmentation – OCR – CNN – Computer Vision – OpenCV – YOLOv3 – YOLOv8 – ByteTrack
AI Frameworks & Libraries: PyTorch – TensorFlow/Keras – Ultralytics – NumPy
Embedded Systems: STM32 – ESP32 – Embedded C – UART – I^2C – SPI
Hardware Design: Altium Designer – SolidWorks – Proteus
Development Tools: Git – GitHub – PyCharm – Visual Studio Code – Jupyter Notebook

LANGUAGES

Arabic: Native proficiency **| English:** Professional Working Proficiency **| French:** Intermediate proficiency

ASSOCIATIVE ACTIVITIES

IEEE ENICarthage Student Branch – Member **| October 2021 – December 2024**
ENICarthage Robots – Head of Line Following Committee **| January 2022**
ENICarthage Internal Contest, Line-Following Competition – 1st Place Winner **| January 2022**